

Exercises

1. Are rewards in reinforcement learning discrete, continuous, or mixed random variables? Try to come up with examples.
2. Let the random variable X be the number of 6's obtained after tossing N dice. What values can X take? Is it discrete or continuous?
3. Throwing a fair coin has a probability of landing heads of 0.5. Construct a random variable with a pmf and show that it is indeed a pmf.

Hint: must sum to 1.

4. **Most of the time, one can swap sums to integrals to switch from the discrete to the continuous case (although not completely correct).

A pdf is a function f such that $f(x) \geq 0$ for all x and

$$\int_{-\infty}^{\infty} f_X(x) dx = 1.$$

Note that this means that its integral exists. A pdf has density $f_X(x)$ at each point.

To get probabilities, we need the area under the curve of the pdf over an interval:

$$\mathbb{P}(a < X < b) = \int_a^b f_X(x) dx,$$

read as the probability X is between a and b .

A continuous uniform distribution has pdf $f(x) = \frac{1}{b-a}$ for $x \in (a, b)$ and $f(x) = 0$ otherwise. Show that it is indeed a pdf.

Hint: the function has to be non-negative and the integral must equal 1.

5. Show that for a random variable X with a pdf $f(x)$, $\mathbb{P}(X = x) = 0$ for all x . To get a probability from a pdf, use

$$\mathbb{P}(a \leq X \leq b) = \int_a^b f(x) dx.$$

Hint: $\mathbb{P}(X = x) = \mathbb{P}(x \leq X \leq x)$.

6. Consider this grid, where each cell is a "room" X_t denoting our position at time t .

1	2
3	4

Assume we are moving randomly and need to move to an adjacent room each time. The probability of being in any room is equiprobable, namely:

$$\mathbb{P}(X_t = 1) = \mathbb{P}(X_t = 2) = \mathbb{P}(X_t = 3) = \mathbb{P}(X_t = 4) = 0.25$$

Now let X_t denote our current room. What are $\mathbb{P}(X_{t+1} = 3 \mid X_t = 3)$, $\mathbb{P}(X_{t+1} = 3 \mid X_t = 2)$, $\mathbb{P}(X_{t+1} = 1 \mid X_t = 3)$, and $\mathbb{P}(X_{t+2} = 1 \mid X_t = 4)$? Remember we can only move to adjacent rooms and not diagonally and cannot stay in the same room.

7. An event is a set of outcomes, e.g. $A = \{\text{we draw a king from a deck of cards}\}$ or $B = \{X = x\}$, where X is a random variable. A, B is also an event that would mean that simultaneously we draw a king from a deck of cards and X takes the value x .

Consider events A and B such that $\mathbb{P}(B) > 0$. Bayes' theorem states that

$$\mathbb{P}(A | B) = \frac{\mathbb{P}(B | A)\mathbb{P}(A)}{\mathbb{P}(B)}.$$

Prove it.

Hint: Write out the conditional probabilities $\mathbb{P}(A | B)$ and $\mathbb{P}(B | A)$ and get an expression for $\mathbb{P}(A, B)$ from the latter.

8. Suppose that two factories supply light bulbs to the market. Factory X 's bulbs work for over 5000 hours in 99% of cases, whereas factory Y 's bulbs work for over 5000 hours in 95% of cases. It is known that factory X supplies 60% of the total bulbs available and Y supplies 40% of the total bulbs available. What is the chance that a purchased bulb will work for longer than 5000 hours?

Hint: Use the law of total probability for events, not for random variables.

*Taken from https://en.wikipedia.org/wiki/Law_of_total_probability

9. Consider the environment:

1	S	-1
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Assume the agent starts in the middle cell (S). If the agent receives a reward, the episode ends. The left cell gives a reward of 1, and the right cell of -1. Let L and R denote the only possible actions, namely left and right.

- What is the expected return of the following policies given the starting state S ?
 - $\pi(L | S) = 0.4$
 - $\pi(L | S) = 0.2$
 - $\pi(R | S) = 0.6$
- The expected value of the return from a policy given the starting state S is 0.5. What is the policy?
- What is the policy that maximizes the expected return? Is it stochastic or deterministic?

10. Consider the discrete random variables X and Y that take values 0 and 1 with a joint pmf:

	$Y=0$	$Y=1$
$X=0$	0.25	0.4
$X=1$	0.3	0.05

Find the marginals $\mathbb{P}(X), \mathbb{P}(Y)$, the probability $\mathbb{P}(X = 1, Y = 0)$, the conditional probability $\mathbb{P}(X = 1 | Y = 0)$ and decide whether X and Y are independent.

11. Is the following function a pmf for a finite action space $\mathcal{A}$ and an action-value function $Q_t(a)$?

$$\pi_t(a) = \begin{cases} (1 - \varepsilon) + \frac{\varepsilon}{|\mathcal{A}|} & \text{if } a = \operatorname{argmax} Q_t(a) \\ \frac{\varepsilon}{|\mathcal{A}|} & \text{otherwise} \end{cases}$$

12. Consider a random student from a class doing the Reinforcement Learning course and the probabilities $\mathbb{P}(\text{the student will pass the course})$ and $\mathbb{P}(\text{the student will study a lot and will pass the course})$. Which one is more likely? Give a mathematical argument.

Hint: The answer might be counterintuitive. Consider $\mathbb{P}(A | B) = \frac{\mathbb{P}(A, B)}{\mathbb{P}(B)}$ and remember that probabilities take values in $[0, 1]$.

13. Show any conditional probability is greater or equal to the joint probability:

$$\mathbb{P}(S | P) \geq \mathbb{P}(S, P).$$

When does this become an equality?

14. Show that the two definitions for independence are equivalent:

$$\mathbb{P}(X, Y) = \mathbb{P}(X) \times \mathbb{P}(Y) \iff \mathbb{P}(X | Y) = \mathbb{P}(X)$$

Hint: For both the if and only if part, use the definition $\mathbb{P}(X | Y) = \frac{\mathbb{P}(X, Y)}{\mathbb{P}(Y)}$.

15. Show that $r(s, a) = \sum_r r \sum_{s'} p(s', r | s, a)$.

Hint: apply the definition of expected value, then use the conversion from the three-argument to the four-argument function by getting the marginal of the reward.

16. Show that if $\mathbb{P}(B, C) > 0$ and $\mathbb{P}(B | C) > 0$, then

$$\frac{\mathbb{P}(A | C)}{\mathbb{P}(B | C)} = \frac{\mathbb{P}(A, C)}{\mathbb{P}(B, C)}.$$

17. Show that $r(s, a, s') = \sum_r r \frac{p(s', r | s, a)}{p(s' | s, a)}$.

Hint: apply the definition of expected value and then the definition of conditional probability, use the previous exercise, and the numerator and denominator should exactly come up to the definitions of the transition functions.

18. Assume the random variable X represents the height of Dutch people (in cm, up to any precision point) and Y represents their assigned sex (F or M). Are $\mathbb{P}(X \in S | Y = F)$ and $\mathbb{P}(X \in S | Y = M)$ independent, where S is some interval of heights? Give an informal argument (in words).

19. State in words if the following converge or diverge. If they converge, find the limit informally (just write its value).

- $a_n = \frac{1+n}{n}$
- $b_n = \frac{1}{n+1}$
- $c_n = \frac{n^2+1}{n+1}$

20. State in words if the following converge or diverge. If they converge, find the limit informally. For the last one, it is enough to say whether it converges or not.

- $\sum_{n=1}^{\infty} 1$
- $\sum_{n=1}^{\infty} n$
- $\sum_{n=1}^{100} 1$
- $\sum_{n=0}^{\infty} \frac{1}{2^n}$
- $\sum_{n=1}^{\infty} \frac{1}{2^n}$
- $\sum_{n=0}^{\infty} \frac{1}{n}$

- $\sum_{n=1}^{\infty} \frac{1}{n^{0.001}}$
- $\sum_{n=0}^{\infty} \frac{1}{n^3}$

21. Compute the following derivatives:

- $\frac{d1}{dx}$
- $\frac{d(3x)}{dx}$
- $\frac{d(5x+5)}{dx}$
- $\frac{d}{dx}(x-2)$

22. Compute the following partial derivatives:

- $\frac{\partial 1}{\partial x}$
- $\frac{\partial(3x+2y)}{\partial x}$
- $\frac{\partial(3x+2y)}{\partial y}$
- $\frac{\partial(3x+2y)}{\partial x \partial y}$
- $\frac{\partial}{\partial x} \mathbb{E}(3x+5y)$
- $\frac{\partial 1}{\partial x} \mathbb{E}(y-5)$

23. Compute the following gradients. Try to push the gradient as much as possible, which should be the final answer. Leave expected values and fractions as such. Note that some examples might already be maximally simplified, and if so, it is enough to just recognize it.

- $\nabla 100$
- $\nabla f(s)$
- $\nabla_x f(s)$
- $\nabla_a \mathbb{E}(\pi(a) + b(s))$
- $\nabla_a \log \mathbb{E}(\pi(a))$

24. Mark as T/F. Explain why.

- $\nabla_{\theta} f(s) \pi_{\theta}(a) = f(s) \nabla_{\theta} \pi_{\theta}(a)$
- $\nabla_s f(s) = f(\nabla_s s)$
- $f(\nabla_x x) = f(1)$
- $\nabla_a (\pi(a) + b(s)) = \nabla_a \pi(a)$
- $\nabla_{\theta} \mathbb{E}[(G - \|\theta\|_2) \log \pi_{\theta}(a)] = \nabla_{\theta} \mathbb{E}[-\|\theta\|_2 \log \pi_{\theta}(a)]$
- $\sum_a b(s) \nabla_a \pi(a|s, \theta) = 0$, assuming swapping the sum and gradient is allowed

25. Show that it is not always the case that one can switch gradients and sums by providing a counterexample.

Hint: Take the sum $1+2$ and express it using notation.

26. Show that for a discrete random variable X with a pmf $\pi(x)$, $\nabla_x \mathbb{E}(\log \pi(X)) = 0$.

Hint: Here swapping the gradient and a sum is correct (if interested, check uniform convergence and derivative of log-likelihood). It is generally allowed to swap the gradient and sum whenever we are working with sums of pmfs, expected values, or sample means (whenever they are finite).

Hint: Push the gradient in, use the definition of expected value, use the properties of the derivative of the natural logarithm, then swap the gradient and the sum, the pmf must sum to 1, the gradient of 1 is 0.

27. Given $f(x) = x^2$, $f'(x) = 2x$, $x_0 = 10$, $f(x_0) = 100$, for what learning rate does gradient descent find the global minimum in one step?

28. Consider $f(x) = x^3 - x^2$ and provide the range of starting points x_0 such that gradient descent will get stuck in a local minimum, assuming a small enough learning rate to guarantee convergence. What happens if $x_0 = 0$?

Hint: No need to compute any derivatives or iterations. Does the global minimum even exist? Use <https://www.desmos.com/calculator> for plotting. Also, any small learning rate will eventually reach the local minimum given a good starting condition.

29. Consider $f(x) = x^4 - x^3$ and provide the range of starting points x_0 such that gradient descent will get stuck in a flat region forever (assuming a small enough learning rate such that GD will converge).

Hint: Plot it in Desmos. Also, $f'(0) = 0$.